

A Comparative Study of Clustering Algorithms: K-means, K-means++, DBSCAN and Spectral Clustering

Liu Zhiyuan 1025040921, Nanjing University of Posts and Telecommunications

Abstract—Clustering is a fundamental task in data mining and big data analysis, aiming to discover hidden structures in unlabeled data. However, the performance of clustering algorithms is highly dependent on the characteristics of the underlying datasets, such as cluster shape, density distribution, noise, and initialization sensitivity. In this project, we conduct a comprehensive comparative study of four representative clustering algorithms: K-means, K-means++, DBSCAN, and Spectral Clustering.

Through experiments on synthetic datasets with different properties, including well-separated spherical clusters, imbalanced clusters, and non-convex structures, we systematically analyze the strengths and limitations of each method. The results show that K-means is computationally efficient but sensitive to initialization and limited to convex cluster shapes, while K-means++ significantly improves stability by optimizing the initialization process. DBSCAN demonstrates strong robustness to noise and excels at identifying arbitrarily shaped clusters without requiring the number of clusters in advance, but its performance is sensitive to parameter selection and varying data densities. Spectral Clustering achieves superior performance on non-convex datasets by leveraging graph-based representations, at the cost of higher computational complexity.

In general, this study highlights that no single clustering algorithm is universally optimal. Instead, the suitability of a clustering method strongly depends on the intrinsic characteristics of the data. The experimental findings provide practical information for selecting appropriate clustering algorithms in real-world big data analysis tasks.

Index Terms—Clustering, K-means, K-means++, DBSCAN, Spectral Clustering, Non-convex Data, Data Characteristics.

I. INTRODUCTION

CLUSTERING is a fundamental task in unsupervised learning and plays a crucial role in big data analysis, exploratory data mining, and pattern recognition. By grouping unlabeled data into meaningful subsets based on similarity, clustering enables the discovery of underlying structures in complex datasets without prior knowledge. As modern applications continuously generate large volumes of heterogeneous data, effective clustering techniques have become increasingly important for tasks such as customer segmentation, anomaly detection, and data summarization.

Despite its wide adoption, clustering remains a challenging problem because real-world data often exhibit diverse and complex characteristics. Different datasets may vary significantly in terms of cluster shape, density distribution, noise level, and scale. Classical clustering algorithms, such as K-means, are computationally efficient and easy to implement but rely on strong assumptions about cluster geometry and initialization, which may limit their applicability in practice.

To address these limitations, alternative approaches have been proposed, including improved initialization strategies (e.g. K-means++), density-based methods (e.g. DBSCAN) and graph-based techniques (e.g. Spectral Clustering). Each of these methods is designed to handle specific data characteristics and therefore exhibits distinct advantages and shortcomings in different scenarios.

Motivated by the absence of a universally optimal clustering algorithm, this project aims to systematically investigate how different clustering methods perform under varying data characteristics. By comparing representative distance-based, density-based, and graph-based clustering algorithms on synthetic datasets with distinct structural properties, this study seeks to provide practical insights into the relationship between data characteristics and algorithm selection. Such an analysis is particularly valuable for guiding the application of clustering techniques in real-world big data analysis tasks, where inappropriate algorithm choices may lead to misleading or suboptimal results.

II. RELATED WORKS

Clustering has been extensively studied in statistics, data mining, and machine learning, with methods typically categorized into distance-based, density-based, graph-based, hierarchical, and model-based approaches. Among the most widely used partitioning methods, K-means dates back to early work by MacQueen [1], who introduced the “k-means” procedure for partitioning multivariate observations, and Lloyd [2], whose formulation is closely related to the modern iterative centroid update algorithm. While K-means is attractive due to its simplicity and computational efficiency, it is well known to be sensitive to initialization and to favor convex (roughly spherical) cluster structures.

To mitigate poor initialization, K-means++ [3] was proposed as a careful seeding strategy that selects initial centers with probabilities proportional to the squared distance from already chosen centers, providing both improved practical stability and theoretical approximation guarantees. In practice, this initialization often reduces the risk of converging to suboptimal local minima, especially when clusters are imbalanced or partially overlapping.

Beyond centroid-based methods, density-based clustering has been developed to address non-convex clusters and noise. A representative algorithm is DBSCAN [4], which defines clusters as dense regions separated by low-density areas and

explicitly labels outliers as noise points. DBSCAN is effective for discovering arbitrarily shaped clusters without requiring the number of clusters in advance, but it may be sensitive to parameter settings (e.g., ϵ and MinPts) and can struggle when cluster densities vary significantly.

Spectral clustering represents a graph-based paradigm: it constructs a similarity graph over data points and partitions the graph using eigenvectors of a graph Laplacian. Foundational work [5] includes normalized cut formulations and practical spectral algorithms [6] that cluster in an embedding space. A widely cited tutorial [7] further clarifies the underlying graph constructions and Laplacian choices. Spectral clustering is particularly effective for non-convex structures but typically incurs higher computational cost due to eigen-decomposition and can be sensitive to graph construction parameters (e.g., the number of neighbors in kNN graphs).

In addition to the algorithms evaluated in this project, several classical and scalable methods are frequently adopted in big data scenarios. Hierarchical clustering, such as the Ward method, builds a dendrogram by greedily merging clusters to optimize an objective, offering interpretability, but often facing higher computational cost for large datasets. To address scalability, BIRCH summarizes large datasets using a clustering feature tree (CF-tree) and performs incremental clustering with linear scalability in practice, though it may depend on summarization quality and parameter choices. For density-based analysis under varying densities, OPTICS [8] provides an ordering that reveals clustering structure across a range of density thresholds, improving upon single- ϵ limitations. Modern extensions such as HDBSCAN further improve the robustness to parameter selection and better handle variable-density clusters by building a hierarchical density-based structure and extracting stable clusters.

Other notable alternatives include Mean Shift, a nonparametric mode-seeking approach that can find arbitrarily shaped clusters but may be computationally expensive and sensitive to bandwidth selection. Affinity Propagation performs message passing to select exemplars without pre-specifying the number of clusters, but can be sensitive to preference parameters and memory usage for large datasets. Finally, model-based clustering (e.g., Gaussian mixture models optimized via the EM algorithm) provides probabilistic assignments and flexibility, but may require careful model selection and can inherit sensitivity to initialization and distributional assumptions.

Overall, prior work suggests that clustering performance is strongly coupled with data characteristics (shape, density variation, noise, and scale). This motivates a comparative evaluation across representative clustering paradigms to better understand practical algorithm selection for different dataset properties.

III. PROBLEM STATEMENT

In this project, the clustering problem is formulated as a data mining task under an unsupervised learning setting. Given a dataset

$$X = \{x_1, x_2, \dots, x_n\}, \quad x_i \in \mathbb{R}^d \quad (1)$$

the objective is to partition the data into a set of clusters such that samples within the same cluster are more similar to each other than to those in different clusters, according to a predefined similarity or distance measure.

Unlike supervised learning, no ground-truth labels are available during the clustering process. As a result, clustering algorithms must rely on intrinsic data properties, such as geometric structure, density distribution, or neighborhood relationships, to infer meaningful groupings. Different clustering paradigms adopt different assumptions about these properties, which directly affects their applicability and performance on real-world datasets.

More specifically, centroid-based methods such as K-means assume that clusters can be represented by their mean vectors and are typically optimized by minimizing the within-cluster sum of squared distances. This assumption implicitly favors convex and roughly spherical clusters. Density-based methods, such as DBSCAN, define clusters as connected regions of high point density and treat sparse regions as separators or noise, enabling the discovery of arbitrarily shaped clusters without requiring the number of clusters in advance. Graph-based approaches, such as Spectral Clustering, construct a similarity graph over data points and seek an optimal partition by analyzing the spectral properties of the graph Laplacian, making them suitable for capturing non-linear and non-convex structures.

The core problem addressed in this study is to understand how these different clustering assumptions interact with varying data characteristics. Specifically, we aim to investigate how cluster shape, density imbalance, noise, and initialization strategies influence the effectiveness of different clustering algorithms. By formalizing clustering as a data mining task with heterogeneous data properties, this project seeks to provide a structured framework for comparing clustering algorithms and guiding practical algorithm selection in big data analysis scenarios.

IV. ALGORITHMS

To address the clustering problem formulated in Section III, this project adopts multiple representative clustering algorithms from different paradigms, including centroid-based, density-based, and graph-based methods. Each algorithm is briefly introduced with its underlying objective function or mathematical formulation, highlighting the assumptions it makes about data structure.

A. K-means Clustering

K-means is a classical centroid-based clustering algorithm that aims to partition a dataset into k clusters by minimizing the within-cluster sum of squared distances. Formally, given a set of cluster centers $\{\mu_1, \mu_2, \dots, \mu_k\}$, K-means solves the following optimization problem:

$$\min_{\{C_j\}_{j=1}^k} \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2, \quad (2)$$

where C_j denotes the set of data points assigned to cluster j , and μ_j is the mean of all points in C_j . The algorithm proceeds iteratively by alternating between two steps:

- 1) Assignment step, where each data point is assigned to the nearest cluster center based on Euclidean distance;
- 2) Update step, where each cluster center is recomputed as the mean of the assigned points.

Despite its simplicity and computational efficiency, K-means is sensitive to the choice of initial cluster centers and implicitly assumes that clusters are convex and roughly spherical in shape [1], [2].

B. K-means++ Initialization

K-means++ is an improved initialization strategy designed to enhance the stability and convergence quality of K-means. Instead of selecting initial centers uniformly at random, K-means++ chooses centers in a probabilistic manner that favors distant points. After selecting the first center uniformly at random, each subsequent center x is chosen with probability:

$$P(x) = \frac{D(x)^2}{\sum_i D(x_i)^2}, \quad (3)$$

where $D(x)$ denotes the distance between point x and the nearest already selected center.

This strategy increases the likelihood that initial centers are well separated, thereby reducing the probability of poor local minima. K-means++ has been shown to provide theoretical approximation guarantees and improved empirical performance, especially on imbalanced datasets [3].

C. DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm that defines clusters as contiguous regions of high point density [4]. Two parameters are required: the neighborhood radius ϵ and the minimum number of points $MinPts$.

For a point x_i , its ϵ -neighborhood is defined as:

$$N_\epsilon(x_i) = \{x_j \mid \|x_j - x_i\| \leq \epsilon\}. \quad (4)$$

A point is classified as a core point if $|N_\epsilon(x_i)| \geq MinPts$. Clusters are formed by recursively expanding density-connected core points, while points that do not belong to any cluster are labeled as noise.

DBSCAN does not require the number of clusters to be specified in advance and is capable of identifying arbitrarily shaped clusters as well as outliers. However, its performance is sensitive to parameter selection and may degrade when cluster densities vary significantly.

D. Spectral Clustering

Spectral Clustering is a graph-based clustering approach that leverages the spectral properties of a similarity graph constructed from the data [5], [6]. Given a dataset, a weighted graph $G = (V, E)$ is built, where each vertex represents a data point and edge weights encode pairwise similarities.

Let W denote the similarity matrix and D the degree matrix with $D_{ii} = \sum_j W_{ij}$. The (normalized) graph Laplacian is defined as:

$$L_{sym} = I - D^{-1/2} W D^{-1/2}. \quad (5)$$

Spectral clustering computes the eigenvectors corresponding to the smallest eigenvalues of L_{sym} and embeds the data into a low-dimensional space spanned by these eigenvectors. A conventional clustering algorithm, such as K-means, is then applied in this spectral embedding space to obtain the final cluster assignments.

By incorporating global graph structure, Spectral Clustering is particularly effective in capturing non-linear and non-convex cluster structures. Nevertheless, it typically incurs higher computational cost due to eigen-decomposition and is sensitive to graph construction parameters [7].

V. EVALUATION

A. K-means and K-means++: Initialization Sensitivity under Cluster Size Imbalance

1) *Ideal Conditions for K-means Clustering:* To establish a baseline for subsequent comparisons, we first examine a scenario in which the underlying assumptions of K-means are well satisfied. The dataset used in this experiment consists of three well-separated clusters in a two-dimensional feature space, each exhibiting an approximately spherical distribution with comparable sample sizes. Such data characteristics closely match the centroid-based modeling assumptions of K-means.

In this setting, K-means is initialized with three centroids sampled from a narrow Gaussian distribution centered near one of the true clusters. Despite this suboptimal and localized initialization, the algorithm converges reliably within a limited number of iterations. As observed in the experiment, convergence is achieved after 19 iterations, indicating that the iterative reassignment and centroid update steps are sufficient to recover the correct cluster structure when clusters are clearly separated.

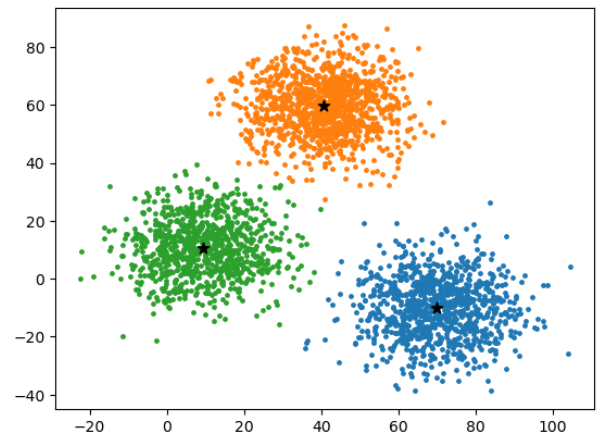


Fig. 1: Clustering result of K-means on a well-separated dataset

As shown in Fig. 1, the final clustering result closely aligns with the intrinsic data distribution, with each cluster accurately

represented by a single centroid. This result demonstrates that, under favorable conditions—namely, well-separated and approximately spherical clusters—K-means exhibits strong robustness to initialization and efficient convergence. From a computational perspective, each iteration requires computing distances between all data points and cluster centroids, resulting in a time complexity of $O(nk)$ per iteration, where n denotes the number of samples and k the number of clusters. The space complexity is dominated by storing the dataset, cluster assignments, and centroids, leading to an overall space complexity of $O(nd+kd)$, which is well suited for large-scale data when cluster structure is simple and well defined.

2) Initialization Sensitivity under Limited Iteration Budgets: As shown in Fig. 2, when K-means is initialized randomly and the number of iterations is explicitly constrained, its convergence behavior becomes highly sensitive to the quality of the initial centroids. Even on generally well-structured datasets with approximately spherical clusters, poor initialization can delay the discovery of meaningful partitions within a limited iteration budget. As shown in the experiments, random initialization requires a substantially larger number of iterations to approach a competitive solution.

Specifically, when the maximum number of iterations is set to 5 or 10, K-means with fixed random initialization produces significantly higher objective values, indicating that the algorithm has not yet stabilized around an optimal configuration. The clustering results at these early stopping points remain partially misaligned with the underlying data structure, as reflected by large inertia values exceeding 2.2×10^6 . Only when the iteration budget is increased to 20 does standard K-means approach a solution comparable in quality to that obtained by more informed initialization strategies.

In contrast, K-means++ demonstrates a markedly different convergence profile. By incorporating distance-aware seeding, K-means++ places initial centroids in well-separated regions of the data space, allowing the algorithm to capture the global structure of the dataset at a much earlier stage. As evidenced by the experiment, K-means++ achieves a substantially lower objective value after a single iteration, reaching an inertia level comparable to that of standard K-means after 20 iterations. This result highlights the advantage of improved initialization in reducing the number of iterations required to reach a high-quality solution.

From a computational perspective, the additional cost introduced by K-means++ during initialization is linear in the number of samples and is negligible compared to the overall iterative cost when the iteration budget is limited. Consequently, under finite iteration constraints, K-means++ offers a more favorable trade-off between solution quality and computational efficiency, even on datasets that do not exhibit extreme imbalance or pathological structures.

3) Impact of Cluster Size Imbalance on K-means and K-means++: To further examine the robustness of centroid-based clustering, we consider a dataset with severely imbalanced cluster sizes, where a large, dense cluster dominates the data distribution and a much smaller cluster is located far away in the feature space. This scenario is particularly challenging for K-means, as the optimization objective is heavily influenced

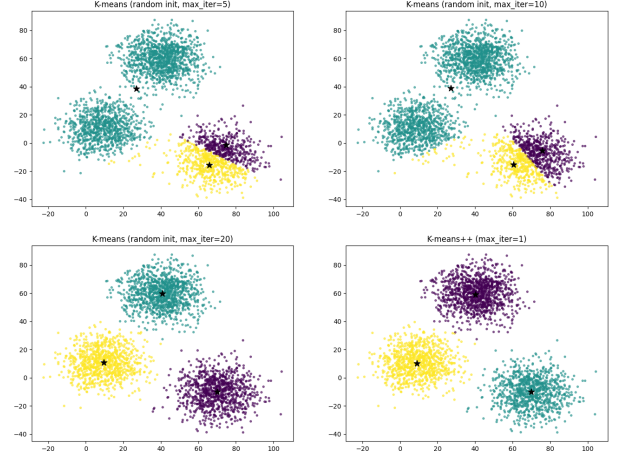


Fig. 2: Clustering results under limited iteration budgets

by the majority cluster.

As illustrated in Fig. 3, even when K-means is allowed to run for a large number of iterations ($max_iter = 200$) with multiple random restarts, the resulting clustering remains suboptimal. The algorithm tends to allocate both centroids to different regions of the majority cluster, effectively partitioning the large cluster into two parts while assigning the minority cluster to the nearest centroid. This behavior leads to a relatively high objective value, indicating that the minority cluster is not properly represented by a dedicated centroid despite extensive iterative refinement.

In contrast, K-means++ exhibits a markedly improved behavior under the same data conditions. By selecting initial centroids based on pairwise distances, K-means++ ensures that one centroid is placed near the distant minority cluster during initialization. As a result, the algorithm is able to capture the true cluster structure almost immediately. Notably, as shown in Fig. 3, K-means++ achieves a significantly lower inertia after only a single iteration ($max_iter = 1$) compared to standard K-means after 200 iterations.

This experiment demonstrates that, under severe cluster size imbalance, increasing the iteration budget alone is insufficient to guarantee satisfactory clustering performance for K-means. Instead, the quality of initialization plays a dominant role. By reducing the likelihood of poor initial centroid placement, K-means++ substantially improves both convergence speed and final solution quality, even under extreme imbalance conditions.

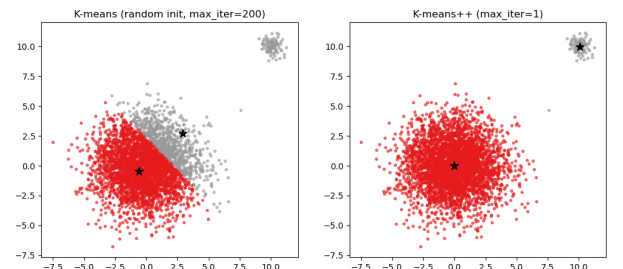


Fig. 3: Clustering results on an imbalanced dataset

B. Clustering Performance on Overlapping Datasets

1) **Overlapping Clusters: K-means versus K-means++:** In this subsection, we investigate the behavior of K-means and K-means++ on datasets with overlapping cluster structures, where the separation between clusters is reduced and cluster boundaries become ambiguous. Specifically, we consider a synthetic dataset consisting of three clusters, in which the two lower clusters partially overlap while the upper cluster remains relatively well separated.

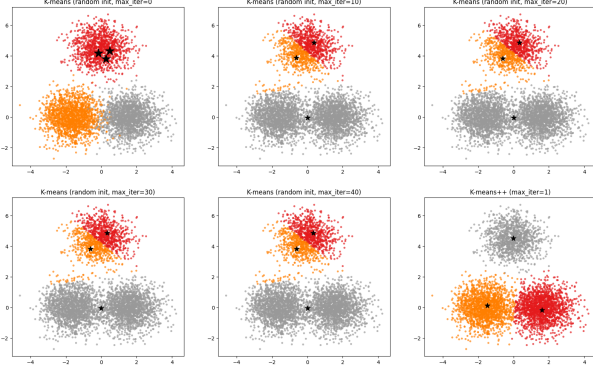


Fig. 4: Clustering results on an overlapping dataset under different iteration budgets.

Fig. 4 illustrates the clustering results obtained by standard K-means with fixed initialization under increasing iteration budgets. When the maximum number of iterations is small, K-means fails to establish stable cluster assignments, and the two overlapping clusters are frequently mixed or incorrectly merged. Even as the iteration budget increases, the algorithm exhibits slow convergence, and the final partition remains sensitive to the initial centroid placement. This behavior highlights the difficulty of relying on iterative refinement alone when cluster boundaries are not clearly defined.

In contrast, K-means++ demonstrates a more robust behavior in the same overlapping setting. By selecting initial centroids that are well separated in the data space, K-means++ is able to capture the global structure of the dataset at an early stage. As shown in Fig. 4, K-means++ achieves a meaningful partition with a single iteration, producing cluster centers that better align with the underlying data distribution. Although the presence of overlap inherently limits the achievable clustering accuracy, improved initialization substantially reduces the number of iterations required to reach a reasonable solution.

These results indicate that, for overlapping clusters, the primary challenge for K-means lies not only in the iterative optimization process but also in the quality of the initial centroids. While increasing the iteration budget can partially mitigate poor initialization, K-means++ provides a more efficient and reliable alternative by addressing this issue directly during the seeding phase.

2) **Application of the Elbow Method on Overlapping Datasets:** The elbow method is a widely used heuristic for selecting the number of clusters in K-means clustering. It is based on analyzing the variation of the within-cluster sum of squared errors (SSE) as a function of the number of clusters

k . In this subsection, we investigate the effectiveness and limitations of the elbow method when applied to datasets with overlapping cluster structures.

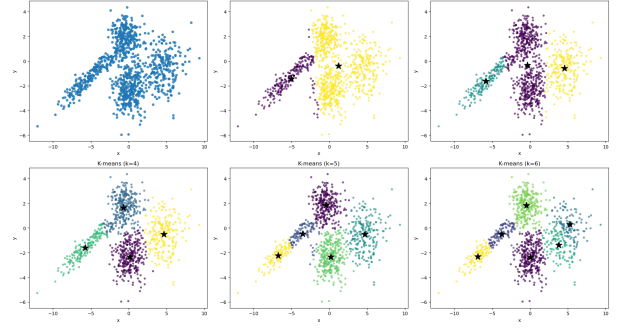


Fig. 5: Clustering results of K-means on an overlapping dataset with different numbers of clusters ($k = 2$ to $k = 6$).

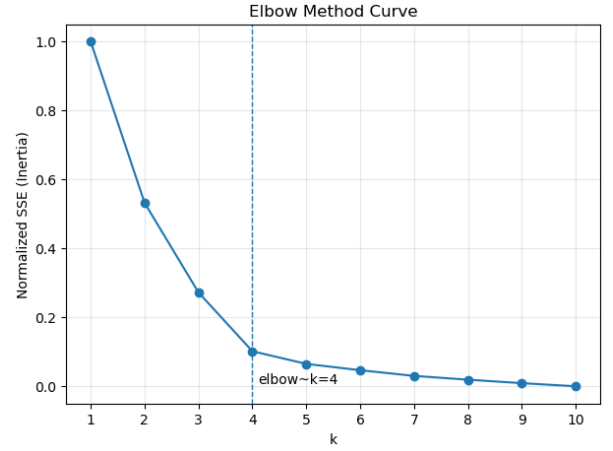


Fig. 6: Elbow method curve showing the normalized sum of squared errors as a function of the number of clusters k .

Fig. 5 presents the clustering results obtained by K-means with different values of k on an overlapping dataset. When k is small (e.g., $k = 2$ or $k = 3$), the algorithm fails to separate the major structures of the data, leading to clear under-segmentation. As k increases to $k = 4$, the main macroscopic groups become distinguishable, and the resulting clusters align more closely with the visually identifiable structures in the data. Further increasing k (e.g., $k = 5$) primarily splits existing clusters into smaller subgroups without introducing new meaningful structures, indicating diminishing returns.

The corresponding normalized SSE curve is shown in Fig. 6. As expected, SSE decreases monotonically with increasing k . However, the rate of decrease exhibits a clear transition: a sharp reduction is observed when increasing k up to $k = 4$, followed by a much slower decline for larger values of $k = 4$. This change in slope corresponds to the so-called “elbow,” which reflects the point at which adding more clusters yields significantly reduced marginal improvement. In this example, $k = 4$ represents a reasonable trade-off between model complexity and clustering quality.

Despite its intuitive appeal, the elbow method exhibits several limitations in the presence of overlapping clusters. First, the elbow point is often not sharply defined, especially when cluster boundaries are ambiguous and the SSE curve becomes smooth. Second, the method relies on subjective visual inspection, which may lead to inconsistent choices across different datasets or observers. Finally, in overlapping scenarios, SSE continues to decrease gradually even beyond the elbow point, as K-means tends to further partition dense regions rather than discover genuinely new clusters. These observations suggest that, while the elbow method can provide useful guidance, it should be complemented with additional validity indices or qualitative analysis when applied to complex, overlapping data distributions.

C. Clustering Performance on Non-Convex Data Structures

1) *Comparison on the Two-Moons Dataset:* To further evaluate the behavior of different clustering algorithms under non-convex data distributions, we consider the two-moons dataset, which is a canonical benchmark featuring intertwined and non-linearly separable cluster structures. As shown in Fig. 7, the dataset violates the implicit assumptions of centroid-based clustering methods.

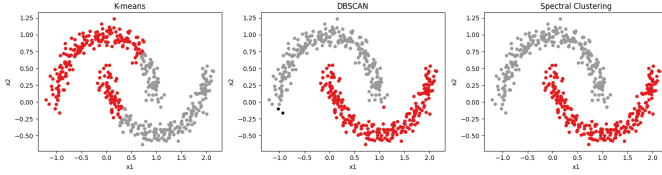


Fig. 7: Clustering results on the two-moons dataset using K-means, DBSCAN, and Spectral Clustering.

K-means partitions the data by assigning points to the nearest centroid in the Euclidean space, which effectively induces linear decision boundaries. Consequently, K-means fails to recover the intrinsic crescent-shaped clusters and instead splits the data into convex regions that do not correspond to the true structure. This limitation is inherent to the algorithm and cannot be resolved by increasing the number of iterations.

DBSCAN, as a density-based method, is capable of identifying arbitrarily shaped clusters and demonstrates improved performance on the two-moons dataset. By grouping points according to local density connectivity, DBSCAN successfully separates the two curved manifolds without requiring the number of clusters in advance. However, its performance is sensitive to the choice of density parameters, such as ϵ and $MinPts$, and may degrade when cluster densities vary or noise levels increase.

Spectral Clustering achieves the most accurate partitioning in this scenario. By constructing a similarity graph and performing clustering in the spectral embedding space, it effectively captures the global manifold structure of the data. As illustrated in Fig. 7, Spectral Clustering correctly separates the two non-convex clusters, demonstrating its advantage over both K-means and DBSCAN on graph-structured data, albeit at a higher computational cost.

2) *Clustering on Concentric Circle Structures:* To further investigate the behavior of clustering algorithms on more challenging non-convex structures, we consider the concentric circles dataset, where clusters are nested and cannot be separated by linear or density-based boundaries. The clustering results obtained by K-means, DBSCAN, and Spectral Clustering are shown in Fig. 8.

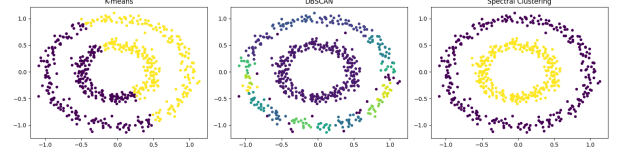


Fig. 8: Clustering results on the concentric circles dataset using K-means, DBSCAN, and Spectral Clustering.

K-means completely fails in this scenario due to its reliance on Euclidean distance to cluster centroids. Since both clusters are centered around the same location, any centroid-based partition inevitably cuts across the circular structures, leading to incorrect assignments regardless of the number of iterations.

DBSCAN also struggles with this dataset. Although it is capable of identifying non-convex clusters, the inner and outer circles exhibit significantly different point densities. As a result, no single choice of the density parameters ϵ and $MinPts$ can simultaneously capture both structures, causing DBSCAN to either fragment clusters or merge them incorrectly.

In contrast, Spectral Clustering successfully separates the concentric circles by exploiting the global connectivity of the data. By constructing a similarity graph and performing clustering in the spectral embedding space, the algorithm effectively transforms the nested non-convex structures into linearly separable representations. This experiment highlights the advantage of graph-based methods in handling complex non-linear cluster geometries, at the cost of increased computational complexity.

VI. CONCLUSION

In this work, we conducted a systematic comparative study of several widely used clustering algorithms, including K-means, K-means++, DBSCAN, and Spectral Clustering, under datasets with different structural characteristics. By designing controlled experiments on well-separated, imbalanced, overlapping, and non-convex datasets, we analyzed both the strengths and inherent limitations of each method.

Our results show that standard K-means is highly efficient and effective when the data satisfy its underlying assumptions, such as spherical and well-separated clusters. However, its performance is strongly influenced by initialization and the choice of the number of clusters, which may lead to suboptimal solutions in practical scenarios. K-means++ significantly improves robustness by introducing a distance-aware initialization strategy, enabling faster convergence and better clustering quality, especially under limited iteration budgets.

For datasets with overlapping or complex geometrical structures, the limitations of centroid-based methods become evident. The elbow method provides a useful heuristic for selecting an appropriate number of clusters, but its effectiveness

depends on the clarity of the SSE curve and may become ambiguous in highly overlapping cases. Density-based clustering, represented by DBSCAN, demonstrates strong capability in discovering arbitrarily shaped clusters and handling noise, but its sensitivity to density parameters limits its applicability when cluster densities vary significantly.

Spectral Clustering consistently achieves superior performance on non-convex datasets by leveraging graph-based representations and global connectivity information. It successfully handles challenging structures such as the two-moons and concentric circles datasets, where both K-means and DBSCAN may fail. This advantage, however, comes at the cost of higher computational and memory complexity, which may restrict its scalability to very large datasets.

Overall, this study highlights that no single clustering algorithm is universally optimal. The choice of an appropriate method should be guided by the intrinsic properties of the data, including cluster shape, density distribution, and scale. The experimental results presented in this work provide practical insights into algorithm selection and demonstrate the importance of aligning clustering techniques with data characteristics in real-world applications.

REFERENCES

- [1] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, 1967, pp. 281–297.
- [2] S. Lloyd, "Least squares quantization in pcm," *IEEE Transactions on Information Theory*, vol. 28, no. 2, pp. 129–137, 1982.
- [3] D. Arthur and S. Vassilvitskii, "k-means++: The advantages of careful seeding," in *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2007, pp. 1027–1035.
- [4] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD)*, 1996, pp. 226–231.
- [5] J. Shi and J. Malik, "Normalized cuts and image segmentation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 22, no. 8, pp. 888–905, 2000.
- [6] A. Y. Ng, M. I. Jordan, and Y. Weiss, "On spectral clustering: Analysis and an algorithm," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2002, pp. 849–856.
- [7] U. von Luxburg, "A tutorial on spectral clustering," *Statistics and Computing*, vol. 17, no. 4, pp. 395–416, 2007.
- [8] M. Ankerst, M. M. Breunig, H.-P. Kriegel, and J. Sander, "Optics: Ordering points to identify the clustering structure," *ACM SIGMOD Record*, vol. 28, no. 2, pp. 49–60, 1999.